

Warudo: Interactive and Accessible Live Performance Capture

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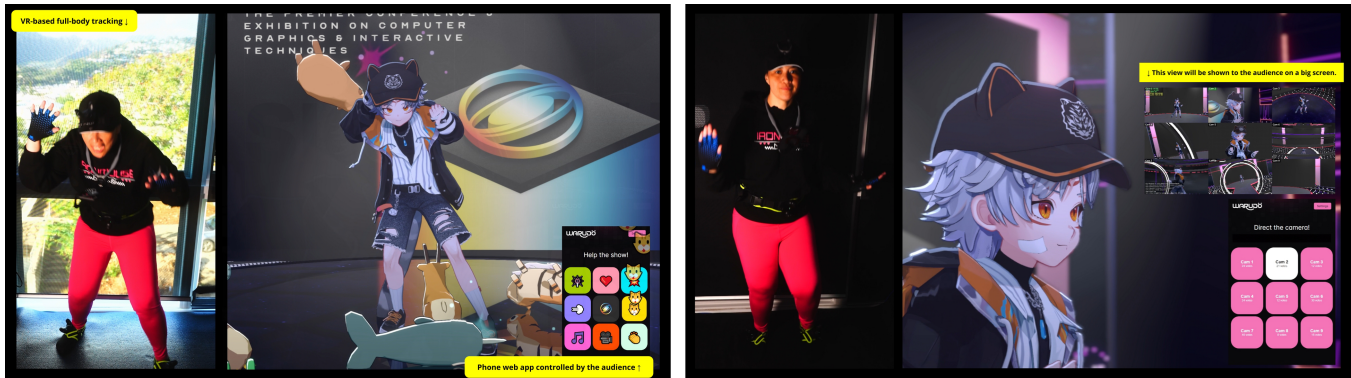


Figure 1: A live, impromptu VTubing performance using Warudo. The audience uses a web app to interact with the performer in real-time, triggering various effects (left) and crowd-directing the cinematography (right).

ABSTRACT

Warudo innovates virtual performances (VTubing) using accessible tracking technologies, procedurally-generated avatar motions, and a visual programming system to customize audience interactions. Its seamless approach enables creators with no technical background to design unique, collaborative live performances. We demonstrate an immersive and real-time VTubing performance driven by the SIGGRAPH audience.

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1 INTRODUCTION

Live interactive performance with virtual avatars, known as “VTubing,” has become increasingly popular, with over 100,000 VTubers worldwide across streaming platforms. Typically, to achieve tracking fidelity and cinematography, VTubing requires an optical tracking system and custom-built software, which are not accessible to most; a one-hour 3D VTubing performance can cost from \$10,000 to

\$50,000, making it a luxury for most creators. Thus, creators often resort to simple webcam-based head tracking applications with few or no audience interactions, limiting the avatar’s expressiveness and interaction with the audience.

Warudo represents a significant leap forward in VTubing technology by providing an integrated platform that enables creators across the spectrum to give professional and captivating VTubing performances. Warudo offers accessible full-body tracking using consumer hardware, including iPhones, webcams, and VR trackers, allowing for a flexible mix-and-match of hardware. For example, an avatar’s facial expressions can be driven by an iPhone, hands and fingers driven by a webcam, while procedurally generated, expressive motions control the rest of the body.

Furthermore, Warudo introduces an intuitive, node-based visual programming system that empowers creators to design unique, participatory experiences which react to live viewer inputs. From effect triggers to crowdsourcing the cinematography, everything is created and previewed in real-time. Creators can utilize Warudo’s Unity and C# SDKs to further expand their setup, integrating custom 3D assets and effects to fit specific needs and artistic visions.

2 TECHNICAL CONTRIBUTIONS

Warudo contributes to the VTubing community with the following technical innovations:

- **Modular design.** Unlike existing VTubing applications that are configuration-based and, thus, limited in flexibility, Warudo separates avatar animation, motion capture, and interaction logic into composable components [2]. Creators can then leverage and combine these “building blocks” to create unique VTubing setups, akin to using a beginner-friendly

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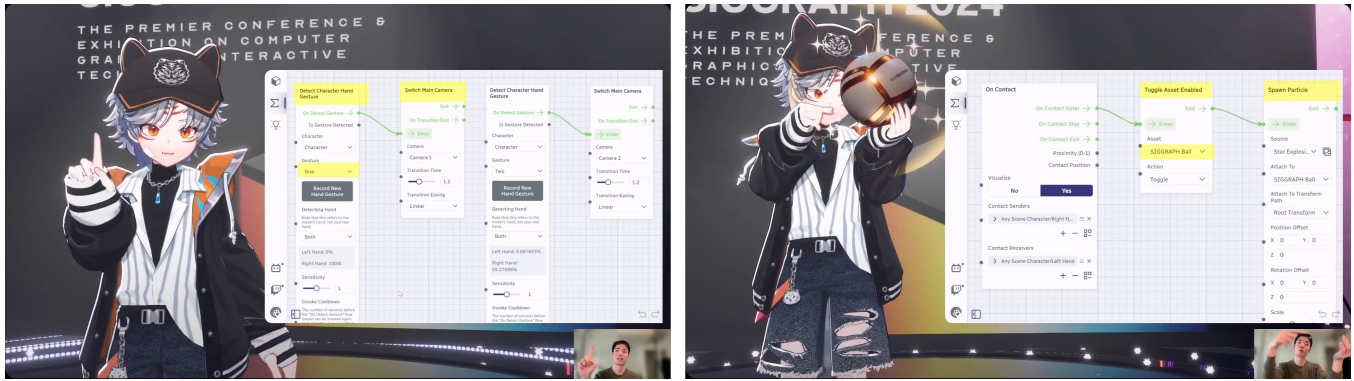


Figure 2: The blueprint system can be used to design interactions without writing code. For example, changing active camera based on performer’s hand gesture (left) or spawning a SIGGRAPH ball when the performer’s hands touch each other (right).

game engine. For example, one can easily create multiple characters with the same tracking data to create a clone army, or combine various budget motion capture systems such as iPhones (face tracking) and Leap Motion controllers (hand tracking) to achieve high-fidelity motion capture. Tracking data can also be sent over the network, enabling remote collaboration and multi-character performances.

- **Visual programming.** One important aspect of VTubing is audience engagement. Warudo’s visual programming (*blueprint*) system [3] allows creators to design complex interactions without writing code: the user simply connects nodes that represent different actions, conditions, and events to create interactive experiences that react to live viewer inputs (Figure 2). The blueprint system can also be used to enhance avatar animations, such as spawning fire breathing effects when the user opens their mouth [4].
- **C# scripting.** Warudo’s scripting system allows one to extend its functionalities by writing C# code that replaces existing behavior or adds new features. This enables creators to implement custom tracking solutions [6], integrate third-party streaming services [5], or create complex interactions [1] that are not possible with the blueprint system.

Together, Warudo eliminates the need for expensive hardware and custom software development, making high-quality 3D VTubing accessible to creators from all backgrounds.

3 LIVE PERFORMANCE

We present Warudo on the *Real-Time Live!* stage to demonstrate its capabilities in live interactive performances with consumer mocap hardware. The audience participates in a real-time, improvised performance by triggering various visual effects and crowd-directing the cinematography using a web app.

- **Introduction.** We import a character model into the virtual environment. We demonstrate the simple, 30-sec VR + iPhone tracking setup process; the character comes to life and starts following the on-stage actor’s movements.
- **Procedurally-generated motions + Mix-and-match mocap.** For those without VR hardware, we show that an iPhone is enough to drive the character’s full body movements. We

also show how different tracking solutions can be combined together.

- **Visual programming.** We explore Warudo’s powerful visual programming by starting with a simple “if pressed a key, then avatar changes face expression” example. Then, we proceed to introduce more blueprints, each with a corresponding interaction effect.
- **Collaborative performance.** We ask the audience to scan QR code with their phone to launch our web app. The audience triggers various effects, including spawning fireworks, throwing animal plushies, playing clapping sounds, etc. The actor then starts dancing, during which the web app becomes an interactive poll, with different camera views listed as options. For every 4 bars in the music, the most voted camera option becomes the active camera; in other words, the audience directs and improvises the show.

4 CONCLUSION

By combining accessible tracking technologies with flexible customization and a community-driven approach, Warudo democratizes the creation of high-quality, animated, and interactive VTubing content. We have made Warudo free for personal and educational use; we envision our novel approach to virtual content creation will also assist and inspire research in interactive storytelling, avatar-mediated education, and social telepresence as a simplified way to implement real-time controlled avatars in a virtual environment.

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